

Erdős Problem #486

EP-486

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $A \subseteq \mathbb{N}$, and for each $n \in A$ choose some $X_n \subseteq \mathbb{Z}/n\mathbb{Z}$. Let $B = \{m \in \mathbb{N} : m \not\equiv 0 \pmod{n} \text{ for all } n \in A \text{ with } m > n\}$. Must B have a logarithmic density, i.e. is it true that $\lim_{x \rightarrow \infty} \frac{1}{\log x} \sum_{\substack{m \in B \\ m < x}} \frac{1}{m}$ exists?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.

Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>